# Mateus Pereira da Silva

## Senior Product Engineer | Full Stack, AI Systems & Distributed Infrastructure

Brazil · Currently in Vancouver, BC, Canada

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## Professional Summary

Product-minded software engineer with **5+ years** shipping production systems across full-stack SaaS, data engineering, infrastructure and applied AI. At VE3, own features for a **30+ customer enterprise platform**, working directly with users and executives; progressed from intern to senior through **three promotions in four years**. Also co-founded and independently built a live travel product, and develop open-source infrastructure.

## Technical Skills

**Languages & product:** Python, Go, TypeScript, JavaScript, SQL, React, Next.js, FastAPI, REST APIs, WebSockets

**Data & distributed systems:** PostgreSQL, Redis, DuckDB, Kafka, Polars, Parquet, Prisma, SAP-originated data integration, pipelines, query tuning

**AI & developer tooling:** LangGraph, LangChain, Pydantic, RAG, pgvector, OpenAI / Anthropic APIs, tool calling, evaluations, JSON Schema

**Cloud & operations:** AWS (Lambda, ECS, S3, RDS, SQS, API Gateway), Docker, Terraform, CI/CD, Cloudflare, Vercel, Prometheus, Grafana

**Testing & security:** pytest, Playwright, Go race tests, GitHub Actions, LangSmith, CloudWatch, Keycloak, JWT/OIDC, production hardening

## Professional Experience

### VE3 | Enterprise SaaS (30+ customers)

Aug 2022 - Present

Senior Product Engineer / Senior Software Engineer | Remote

- **Own product areas end to end**, from discovery with the CEO and customers to architecture, data models, frontend/backend delivery, deployment and production support.
- **Built and operate a production multi-tenant AI analytics assistant** with LangGraph orchestration, 7 domain specialists, ~25 tools and Pydantic structured outputs; added selective domain retries and human clarification.
- **Implemented RAG and guarded text-to-SQL** using OpenAI embeddings, pgvector, DuckDB over S3 Parquet, PostgreSQL and sqlglot AST constraints; maintain tenant-aware context and conversation memory.
- **Built AI evaluation and observability** with source-grounded numerical checks, real-data A/B comparisons, conversation replay, LangSmith tracing and 459 offline tests with mocked LLM calls.
- **Built forecasting and planning systems** spanning 25+ predictive models over millions of records; developed Python/FastAPI and Go services, Polars workers and S3-based data pipelines.
- **Integrated SAP-originated enterprise data** into S&OP/S&OE models and built spreadsheet-grade planning interfaces for customers accustomed to working in Excel.
- **Improved a high-traffic workflow by 75%** through PostgreSQL optimization; separately reduced query cost in another dashboard area by approximately **50x**.
- **Owned infrastructure and reliability** across AWS, Docker and Terraform; built custom Prometheus/Grafana monitoring and alerts, and maintained approximately 3 years without unplanned production downtime.
- **Managed production changes** with staged database releases, SQL reconciliation and recovery; supported a Keycloak/JWT-OIDC authentication migration and remediated external pentest findings.

### Self-Employed | Software Developer

Feb 2021 - Aug 2022

- Built web applications, REST APIs, databases and business automation for clients, managing requirements, implementation, deployment and communication.

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Selected projects & engineering work

## Selected Projects

### Plan Patagonia | Co-Founder & Lead Product Engineer

planpatagonia.com | Next.js 16, React 19, TypeScript, Prisma 7, Neon PostgreSQL, Vercel, Cloudflare R2 | Live in production

- **Co-founded and independently built the digital product from zero to launch**, owning UX, architecture, full-stack implementation, infrastructure and product direction for a Patagonia trip-planning platform.
- **Built a multi-tenant partner directory** with ~38 Prisma models, organizations, invitations, content administration, bulk imports, partner attribution and first-party lead analytics.
- **Shipped a weather-probability engine** built on ~35 years of ERA5 reanalysis data, historical ingestion, precomputed climatology and goal-specific travel conditions.
- **Designed a pre-registered evaluation protocol** and leave-one-year-out backtesting with Brier scores, reliability checks and uncertainty bounds; excluded unverified weather claims from launch.
- **Delivered internationalized and cost-aware infrastructure**, with English/Spanish/Portuguese, SEO, Cloudflare R2 image delivery and scale-to-zero Postgres.

### Workflow Execution Engine (WEE) | Independent Engineer

Go, React, TypeScript, React Flow, WebSockets, JSON Schema, OpenAI / Anthropic APIs, Docker | Local/self-hosted, in development

- **Built a Go workflow runtime** for auditable AI and deterministic engineering processes, with DAG scheduling, bounded concurrency, conditional branches and distinct retry policies.
- **Implemented resilient execution state** using persisted snapshots, replay, restart recovery, retry-from-node and content-addressed artifacts/cache with hash-chained event logs.
- **Added structured-output and cost controls** through JSON Schema contracts, provider adapters, token accounting and execution budgets that carry across resumed runs.
- **Built a React Flow visual editor and live execution UI** with WebSocket events, artifact inspectors, versioned workflow editing and per-node metrics; exposed CLI and Go SDK interfaces.
- **Implemented human approval checkpoints** for write, shell and Git operations plus path, command and secret restrictions; built a 21-node code-review workflow (7 model workers, 14 tool nodes).

### Kafka Go SDK | Open Source

github.com/thettteus/go-kafka-sdk | Go, Apache Kafka, confluent-kafka-go/librdkafka, GitHub Actions | MIT-licensed pre-1.0 module

- **Published a thin, opinionated Go SDK** that reduces producer/consumer boilerplate with one Message type, context-controlled consumption, async delivery reports and a goroutine consumer pool.
- **Added broker-backed integration tests and CI** with a Kafka KRaft container, lint, build and Go race detector; shipped tagged pre-1.0 versions through v0.1.4.

## Additional Engineering Work

- **Sell-Out MPS:** Go API + Python/Polars processing service with S3 bronze/silver datasets and a curated database layer consumed by the existing planning platform.
- **Product judgment:** designed configurable price-prioritization behavior around stakeholder needs, migration risk and fallback rules rather than adding isolated price levels.

## Languages

Portuguese (Native) | English (C1, Advanced) | Spanish (Beginner)